

Baadhi — mapping flood damage from space

IIT Mandi Multimodal AI Hackathon 2026 · Track B · Case study: Trishuli glacial-lake outburst flood, 26 August 2026 (Copernicus EMSR927)

Given any area and a flood date, Baadhi builds a map of flood water and debris from Sentinel-1 radar and Sentinel-2 optical images, counts the buildings, roads and bridges in its path on the pre-event OpenStreetMap, and finds the settlements that lost their road link to a hospital or town — from raw data, in about 3–5 minutes on a laptop — and writes a one-page situation report. Its AI component is a flood segmentation network trained from scratch on Kuro Siwo; on live Sentinel-1 scenes of the 2025 Punjab floods it reaches a mean flood F1 of **0.80** against 0.52 for a radar threshold rule. On the Trishuli case study the flood/debris map agrees with the Copernicus EMS delineation at F1 0.89 (development areas) and **0.86 on the held-out area**.

1 · Problem and what the system answers

A glacial-lake outburst or monsoon flood destroys roads and bridges in narrow valleys within hours; responders need to know **where** the water and debris went, **what** it hit, and **who** is now cut off from care. The system must do this for an area and date chosen live by the judges, using only open satellite data and the map as it was before the event.

Question	Evidence	Output
Where did flood and debris hit?	Sentinel-1 before/after on every usable orbit track; Sentinel-2 when clear; flood model; DEM terrain rules	classes <i>flood water · debris/mud/scoured ground · affected channel</i> , confidence
What was damaged?	overlay on OpenStreetMap at <i>event – 2 days, 00:00 UTC</i>	buildings hit / at the edge, blocked road stretches, bridges over an affected reach, health facilities
Who is cut off?	road graph before vs after (network 25 km beyond the area), walking paths separately	per settlement: minutes to the nearest hospital/clinic and town, before and after
Bonus: where would a flood go?	steepest descent on Copernicus DEM, continued across windows	path and settlements, km downstream, height above the river

2 · Data and the rules we keep

Inputs: Sentinel-1 RTC (γ^0), Sentinel-2 L2A and Copernicus DEM GLO-30/90 from Microsoft Planetary Computer; OpenStreetMap as it was before the event (Overpass "attic" query, two days before the event date at 00:00 UTC — EMSR927 began on 25 Aug 22:00 UTC); Kuro Siwo for training only. **Copernicus EMS, UNOSAT and other published flood maps are never inputs:** EMSR927 and EMSR838 are used only to score results, in code under `experiments/` that the system never imports. No detector threshold was fitted to EMS maps (section 7).

3 · Method

3.1 Flood and debris map

- **Radar change** — for each track, up to three before-images and the first after-image; 3×3 multilook; log-ratio divided by each pixel's own pass-to-pass variability ($z \geq 3$, or two tracks ≥ 2). Layover, shadow and strong foreshortening are masked per track from the DEM and the satellite's heading and incidence.
- **Radar water** — very dark after ($VV \leq -15$ dB, $VH \leq -22$ dB) and ≥ 3 dB darker than before.
- **Optical** — clear pixels only (scene-classification mask), per-date clear-fraction selection, first-clear-pixel composites: vegetation stripped (NDVI -0.20), fresh sediment (bare-soil or brightness rise with NDVI loss), new water (MNDWI). Where Sentinel-2 saw the ground clearly and unchanged, radar change alone does not count (farmland harvests make radar noisy).
- **Terrain** — height above the main river (HAND, pysheds on the DEM with OSM rivers burned in) and slope gate what can be water (≤ 30 m) or debris (≤ 80 m; steep scoured banks allowed within 60 m of the river). Zones must connect to the river corridor; valley floor below the observed flood level is filled where clouds or radar shadow hid the ground.
- **Flood model** — water the network finds (probability ≥ 0.5) enters directly and is not forced to touch the corridor (on plains, flood water breaks into patches between dry fields).

3.2 AI component: flood segmentation model

U-Net with a ResNet-18 encoder (12.5 M parameters), **trained from scratch** on Kuro Siwo only — no ImageNet weights, so no data outside the allowed lists — on a laptop CPU. Input: 8 channels (VV and VH after and before, their differences, slope, height above the local valley floor); output: no water / permanent water / flood water. Official Kuro Siwo event splits: 23 training events, 11 validation events (four training events from different climates moved to validation), **official test events never seen** (including Nepal, 1111007). Loss: weighted cross-entropy + Dice on flood; random crops, rotations, flips, independent calibration offsets on the before/after images; weight averaging (EMA).

Making live data look like the training data. Kuro Siwo is σ^0 with a Lee filter; Planetary Computer gives terrain-flattened γ^0 without filtering. Live images are converted ($\sigma^0 \approx \gamma^0 \cdot \cos \theta$, incidence from the swath geometry) and Lee-filtered, then the *same* feature code as in training runs (`baadhi/mL/features.py`). Inference uses 512-pixel windows blended at the seams, averaged with the mirror image; per pixel, the highest probability among tracks that see it properly. The model is exported to ONNX and run with `onnxruntime` (identical output, 1.6× faster than PyTorch; the dashboard does not need PyTorch).

3.3 Damage, access and flood path

Buildings are rasterised on the 10 m grid: $\geq 50\%$ inside the footprint = *hit*, touching it = *at the edge*. A road is *blocked* where ≥ 20 m of it runs continuously under water or debris; bridges within 30 m of an affected reach are *at risk*. Access: an undirected road graph with class speeds (trunk 50 km/h ... track 12 km/h), multi-source Dijkstra from all hospitals/clinics and towns (including those up to 25 km outside the area) before and after removing blocked stretches and at-risk bridges; a settlement is *cut off* if it loses its route, a *long detour* if the time doubles and grows ≥ 30 min. Footpaths are walked at 3 km/h to tell whether a cut-off village can still be reached on foot. The flood path follows D8 directions on GLO-90 and lists named places ≤ 400 m from the path and ≤ 60 m above the river.

4 • Results — Trishuli GLOF vs Copernicus EMSR927

EMSR927 areas 1–3 were used while developing the detector; **area 5 (Phosretar) was held out and evaluated once**. Scores are computed inside each EMS area of interest, excluding what EMS marked "not analysed".

Area	Precision	Recall	F1	IoU	EMS damaged buildings we flag	Damaged road length in footprint	Damaged bridges flagged / in OSM
AOI 1–2 Syabru Bensi– Timure dev	0.99	0.81	0.89	0.81	95 %	85 %	5 / 5
AOI 3 Bidur dev	0.97	0.88	0.92	0.85	94 %	84 %	20 / 20
AOI 5 Phosretar held out	0.97	0.77	0.86	0.75	63 %	55 %	24 / 30

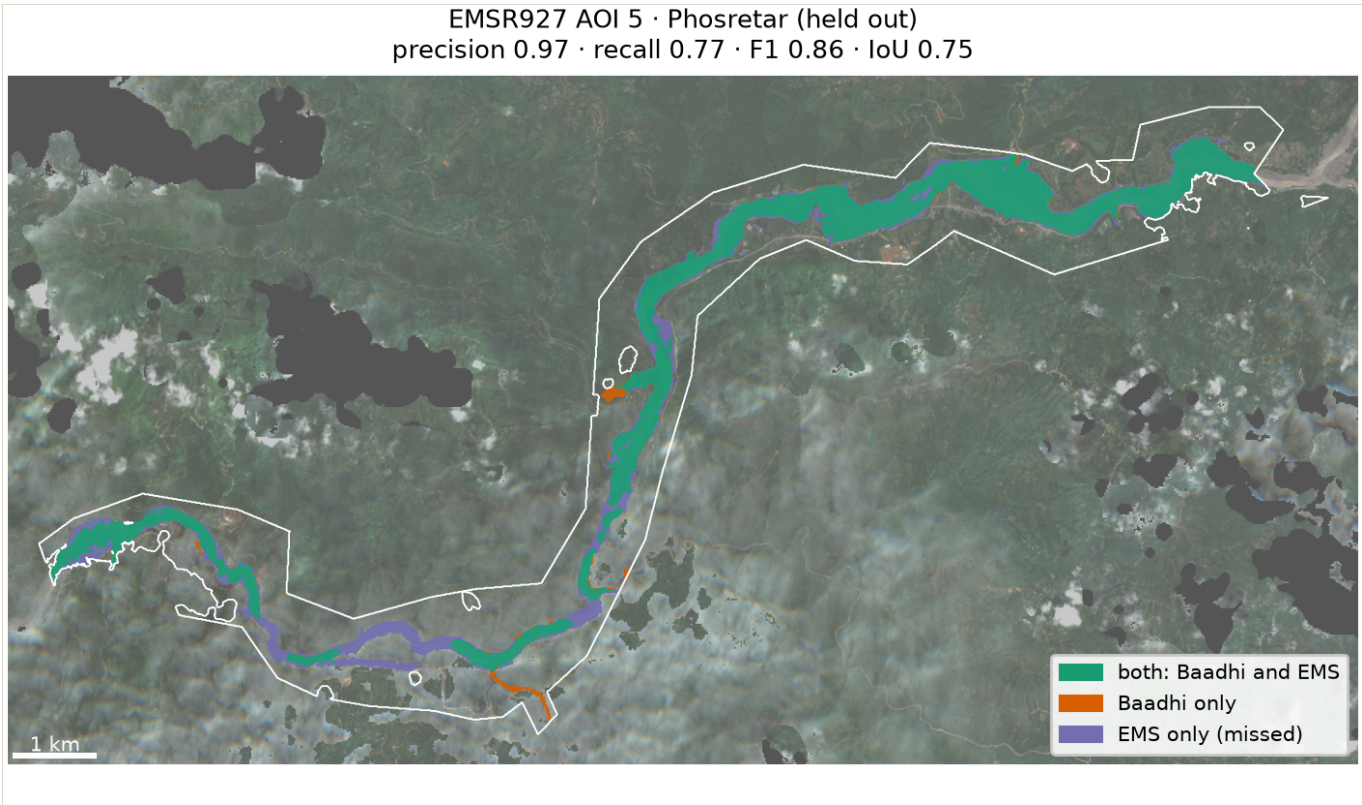


Figure 1. Held-out area: green = both Baadhi and EMS, orange = Baadhi only, purple = EMS only. The missed ground was seen clearly by Sentinel-2 (100 %) and by radar (99 %); it lies higher on the banks — median 11 m above the river on 16° slopes, against 3 m and 3° where both agree — where the flood's upper trace left little change visible at 10 m.

Buildings: in the upper valley EMS graded 760 buildings damaged or destroyed; 95 % of them lie in or at the edge of our footprint. "Precision" of building flags (our hit buildings with an EMS-graded point within 15 m) is 84 % upper, 86 % Bidur, 65 % Phosretar. **Access:** in the upper valley 22 settlements lose their road link to the nearest hospital or town — Syabru Bensi had a clinic 24 minutes away before the event and none by road after. [Step 4 cross-check against EMS road grading: pending]

5 • Results — the flood model on data it never saw

Live Sentinel-1, EMSR838 Punjab floods (Aug 2025). For each EMS area we take the ~14 km box with the most mapped flooding, use exactly the Sentinel-1 pass EMS mapped from, before-images from before the flood began, and score inside the EMS area only.

EMSR838 area	EMS flood	Rule F1	Model F1
AOI01 Rasoo Nagar	143 km ²	0.63	0.95
AOI02 Sharaqpur	65 km ²	0.31	0.91
AOI04 Najabat	26 km ²	0.85	0.90
AOI05 Trimmu	9 km ²	0.30	0.46
Mean		0.52	0.80

Trimmu is a braided river: 317 flood patches with a median of 80 m², many at the scale of a pixel; the monsoon before-images already held water in many braids, which the model then calls permanent rather than new flood water.

6 · Dashboard, report and speed

The dashboard (FastAPI + MapLibre, python `serve.py`) takes an area drawn on the map and a date, streams progress, and shows the classes, confidence and model probability, Sentinel-2 before/after and a radar change composite, buildings, blocked roads, bridges, health facilities and settlements by access status; clicking a settlement shows its travel times. Each run produces the one-page situation report (HTML/PDF, every number from the maps), GeoTIFFs and GeoJSON layers. A 150–200 km² Himalayan valley runs end to end in about 3–5 minutes on a 10-core laptop CPU without a GPU; downloads dominate. [cold-run timings of the practice events: pending]

7 · Limitations and honesty

- **Revisit gaps:** 6–12 days per track; water that drained before the first pass after the event is not seen; nothing is predicted in advance.
- **Radar blind spots** on steep slopes facing towards/away from the satellite; optical images need clear sky (rare in the monsoon).
- **"Hit" is not "destroyed":** at 10 m a building inside the footprint may stand; EMS grades damage from very-high-resolution imagery.
- **OpenStreetMap completeness:** unmapped buildings, bridges and villages are not counted — 9 of the 39 bridges EMS graded at Phosretar are not in OSM.
- **Travel times** use typical class speeds; roads outside the analysed area are assumed passable.
- **Where things were tuned:** detector thresholds are physical and fixed; they were checked on EMSR927 areas 1–3 (development), area 5 was evaluated once. On EMSR838 the model alone is an independent test; two plains rules were added after inspecting AOI01 (model water need not touch the river corridor; unknown height-above-river accepted on flat ground), so the combined detector's EMSR838 scores are a development check.
- **Kuro Siwo's Nepal event lies in the Koshi plains;** mountain performance is shown by the EMSR927 case study.
- Educational prototype, not an operational tool. No images of victims are used.

Attribution

Contains modified Copernicus Sentinel data 2025–2026, processed by Microsoft Planetary Computer. Copernicus DEM: produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors (ODbL), pre-event snapshots via the Overpass API. Kuro Siwo — N. I. Bountos et al., NeurIPS 2024 Datasets and Benchmarks (CC BY 4.0). Checking only: Copernicus Emergency Management Service © European Union (EMSR927, EMSR838). Basemap © OpenFreeMap © OpenMapTiles. MapLibre GL JS (BSD-3-Clause).

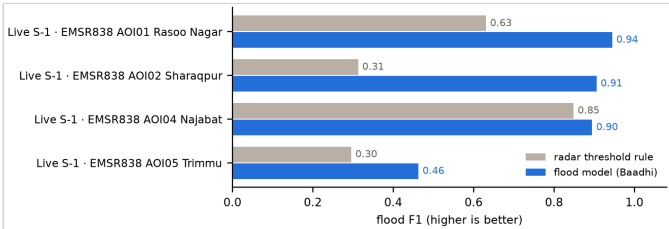


Figure 2. Flood F1 of the model and of the radar threshold rule on scenes never used in training.

Kuro Siwo official test events [final model: pending]. Early checkpoint: Nepal (1111007, Koshi plains) F1 0.76 vs 0.63 for the rule.